

# Metrics

ar005 · 14 May 2026

This page is the reference for every number and every panel the scope shows. It is written to stand on its own: if this is the first page you read, you should come away understanding what the scope is showing, what each metric measures, and what a “healthy” value looks like.

## model

*What it is:* The model variant driving the run. These sit on a ladder from simplest (snnTorch, a standard feedforward LIF network) to most biophysical (PING, with excitatory and inhibitory populations coupled into a gamma-producing loop). See COBANet for the conductance-based model.

*How it’s calculated:* Read from the `–model` CLI flag. A spinner character prefixes the name on live runs so the eye can tell a frozen frame from one that’s still updating.

*Common values:* `snnTorch`, `CUBA`, `COBA`, `PING`.

## dt

*What it is:* The integration timestep in ms — how much real time elapses per simulation step. Smaller  $\Delta t$  is more accurate but more expensive (more steps per trial).

*How it’s calculated:* Read from `–dt`. Held fixed across a training run; can be varied at inference to probe  $\Delta t$ -stability.

*Common values:* 0.1 ms (fine), 0.25 ms (default), 1.0 ms (coarse). PING breaks down above about 1 ms because the excitatory-inhibitory feedback loop can’t complete within a single step.

## N

*What it is:* Number of excitatory neurons in the hidden layer. For PING, the inhibitory population is sized at  $N_E/4$  (the classical 4:1 cortical ratio).

*How it’s calculated:* Read from `–n-hidden`.

*Common values:* 256, 512, 1024.

## in

*What it is:* The peak firing rate of the input neurons, in Hz. An MNIST pixel with intensity 1.0 fires at this rate; a dark pixel fires at zero. Controls how strongly the network is driven.

*How it’s calculated:* Each input neuron emits a Bernoulli spike at each step with probability

$$p = r \cdot \Delta t / 1000$$

where  $r$  is this per-pixel rate (Hz) scaled by pixel intensity. See Training § Input encoding for the full scheme.

*Common values:* 50 Hz for PING, 100 Hz for simpler models. - in the header when not set.

## E

*What it is:* Firing rate of the excitatory population, averaged over all E neurons and over the trial, in Hz. The single most important live readout of network activity.

*How it's calculated:* Let  $s_E \in \{0, 1\}^{T \times N_E}$  be the binary E spike array ( $T$  timesteps,  $N_E$  neurons; entry is 1 if neuron  $i$  spiked at step  $t$ ), and  $T_{\text{sec}} = T\Delta t/1000$  the trial length in seconds. Then

$$r_E = \frac{\sum_{t,i} s_{E[t,i]}}{N_E \cdot T_{\text{sec}}}$$

*Common values:* 1–30 Hz is sparse-coding territory (healthy for snnTorch when classifying MNIST); 40–80 Hz is the gamma-locked regime (PING target); over  $\sim 150$  Hz means the network has saturated; 0 Hz means it's silent.

## I

*What it is:* Firing rate of the inhibitory population, same definition as  $E$ . Only meaningful for models that have an I population (COBA, PING); shows - otherwise.

*How it's calculated:* Same as  $E$ , on the I spike array  $s_I$  and with  $N_I$  inhibitory neurons:

$$r_I = \frac{\sum_{t,i} s_{I[t,i]}}{N_I \cdot T_{\text{sec}}}$$

*Common values:* Typically 2–4 $\times$  the E rate under balanced PING operation — roughly 80–200 Hz.

## f<sub>0</sub>

*What it is:* The dominant oscillation frequency of the E population, in Hz. If the network is gamma-locked the rate signal has a clear peak in its Fourier spectrum;  $f_0$  is the frequency of that peak.

*How it's calculated:* The E spike raster is binned at 2 ms, the resulting time series  $r_E(t)$  is mean-subtracted and FFTed. A peak search runs in the 5–80 Hz band and requires the peak to be at least 3 $\times$  the band median (an SNR gate). A subharmonic check follows: if there is clear power at half the peak frequency ( $\geq 30$  % of the peak's power),  $f_0$  is set to that lower frequency instead — this catches the case where pyramidal neurons fire every second cycle.

*Common values:* 0 Hz (non-oscillating; always for non-PING models, which short-circuit the calculation); 30–80 Hz (gamma, the PING target); anything below 30 Hz is sub-gamma beta. Shows - in the header when 0.

## CV

*What it is:* How bursty the E population's firing is. Takes the total E spike count in each 2 ms bin and computes the coefficient of variation (std / mean) of those counts across bins.

*How it's calculated:* Bin  $s_E$  at 2 ms; let  $n_b$  be the total E spike count in bin  $b$ . Then

$$\text{CV} = \frac{\sigma(n_b)}{\max(\mu(n_b), 10^{-9})}$$

*Common values:* Around 1 for Poisson-like asynchronous firing; well above 1 (often  $\gtrsim 2$ ) for gamma-locked PING, where spikes cluster into cycle peaks and leave the troughs empty; below  $\sim 0.3$  for a clock-like or saturated network.

## **I/E**

*What it is:* The ratio of inhibitory to excitatory population rates. A single number that summarises E–I balance.

*How it's calculated:*

$$I/E = \frac{r_I}{\max(r_E, 10^{-9})}$$

*Common values:* 2–4 under balanced PING operation. Much higher means inhibition has run away; much lower means inhibition is failing or E has saturated.

## **act**

*What it is:* The fraction of excitatory neurons that spiked at least once during the trial. Complements the population rate: a high  $E$  could come from a few neurons firing a lot, or many neurons firing a little — *act* tells you which.

*How it's calculated:*

$$\text{act} = \frac{\#\{i : \sum_t s_{E[t,i]} > 0\}}{N_E}$$

*Common values:* 1–3 % is healthy sparse coding (a classical snnTorch regime). 80 %+ with CV near 0.4 is healthy gamma-locked PING. Persistent values below 1 % or above 95 % alongside stalled accuracy are pathological.

## **e\_raster**

*What it is:* The excitatory spike raster over the trial. Each dot is one spike; the y-axis is E-neuron index, the x-axis is time. This is the primary dynamical read and occupies the largest cell in the grid.

*How to read it:* Vertical bands (near-synchronous firing across the population, at regular intervals) mean gamma-locked PING. A uniformly-speckled cloud means asynchronous firing — normal for CUBA or snnTorch. An empty raster means the network is silent; a solid block means it is saturated.

## **i\_raster**

*What it is:* Same idea as *e\_raster* but for the inhibitory population. Empty for models without an I population.

*How to read it:* Under balanced PING the I raster is denser and at higher rate than the E raster, matching the  $2\text{--}4\times$  I/E ratio.

## **drive**

*What it is:* The mean input current (conductance) reaching the membrane, overlaid with the voltage thresholds a neuron needs to reach to spike.

*How it's calculated:* The input conductance is passed through an exponential synapse with time constant  $\tau_{\text{AMPA}}$  to get the steady-state drive

$$g_{\text{ss}} = \frac{\bar{g}}{1 - e^{-\Delta t / \tau_{\text{AMPA}}}}$$

The bare spike threshold — the drive a neuron needs in the absence of inhibition — is

$$g_{\text{thresh}} = \frac{g_L(V_{\text{th}} - E_L)}{E_e - V_{\text{th}}}$$

The effective threshold adds an inhibition-proportional penalty  $g_i \cdot |E_i - V_{\text{th}}| / (E_e - V_{\text{th}})$ .

*How to read it:* The shaded region is where drive exceeds the effective threshold — that's when E neurons can fire. If drive never crosses threshold the network is dead by construction; if drive is always above, it is saturated.

## weights

*What it is:* Histograms of the current weight values, one sub-histogram per weight matrix (input-to-hidden, hidden-to-output, and any recurrent E-E / E-I / I-E matrices).

*How to read it:* Signed weights (standard SNN) look roughly Gaussian around zero. Dale's-law weights are non-negative so they form a half-normal clamped at zero. A long tail means a few synapses dominate; a narrow stack at zero means many weights have died under regularisation.

## output

*What it is:* The readout logits for each output class, over the course of the trial.

*How it's calculated:* The readout accumulates hidden spikes across time and projects them through a trained linear layer:

$$\hat{y}_t = \left( \sum_{s \leq t} s_s^{\text{hid}} \right) W_{\text{out}} + b_{\text{out}}$$

The argmax at the final step is the model's prediction. See Training § Readout.

*How to read it:* For MNIST there are ten traces; if the network has learned, one trace pulls away from the pack as the trial progresses. If all traces stay flat, or one dominates from  $t=0$ , the run is pathological.

## psd

*What it is:* The power spectral density (frequency content) of the E population rate  $r_E(t)$ . The gamma band (30–80 Hz) is shaded.

*How it's calculated:*  $r_E(t)$  is computed in 2 ms bins over the stimulus window, mean-centred, FFTed, and normalised:

$$\text{PSD}(f) = \frac{|\text{FFT}(r_E - \bar{r}_E)(f)|^2}{\max_f |\text{FFT}(r_E - \bar{r}_E)(f)|^2}$$

*How to read it:* A flat PSD means asynchronous firing. A peak inside the shaded band means gamma. A peak below 30 Hz means sub-gamma (beta). A peak above 80 Hz is rejected by the  $f_0$  search.

## participation

*What it is:* The distribution of per-neuron spike counts across the E population for this trial.

*How to read it:* A stack at zero indicates dead neurons; a stack at the per-neuron ceiling indicates saturated neurons. Both can be invisible from the population-level  $E$  token, which averages across the group.

## acc\_curve

*What it is:* Test accuracy, training loss, and learning rate as a function of epoch — the standard training-progress panel. All three are normalised to their running maximum and plotted on a single 0–100 % axis. Rendered only during training runs.

*How to read it:* A healthy run has accuracy climbing from  $\sim 10$  % (chance on MNIST) toward 85–98 %, and loss dropping monotonically. A flat accuracy with drifting loss usually means the readout can't separate classes.

## rate\_curve

*What it is:* The E and I firing rates as a function of training epoch. Training-only.

*How to read it:* Stable rates within the healthy regime for the model (see *act* above) means the network is holding its operating point as it learns. Monotonic drift toward 0 or toward a ceiling usually means gradient pressure is pushing the network out of its regime.

## grad\_flow

*What it is:* For each weight matrix, the ratio of gradient norm to weight norm — how hard each layer is being updated relative to its current size. Training-only; log y-axis.

*How to read it:* The shaded band from  $10^{-3}$  to  $10^{-2}$  is the healthy window. Values below mean gradients are vanishing (that layer is not learning); above means they are exploding.

## digit\_image

*What it is:* The MNIST digit currently being presented to the network. A small greyscale inset that lets you sync the readout in *output* to the actual input class.

## sweep

*What it is:* A generic progress indicator for parameter-sweep runs. Shows the full sweep range faintly and the completed prefix solidly, with a dot on the current value.

## sweep\_rates

*What it is:* E and I firing rates accumulated across an inference sweep — one point per sweep frame, updated as the sweep progresses. Sweep-only.

*How to read it:* If the trace is flat across the swept variable, the network is stable under that variable. If it fans out by one to two orders of magnitude, it is not — this is the signature read off in a  $\Delta t$ -stability experiment.

## sweep\_f0

*What it is:* Peak oscillation frequency  $f_0$  across the sweep, with the gamma band (30–80 Hz) shaded. Sweep-only.

*How to read it:* A PING run with a robust E→I→E loop holds  $f_0$  inside the shaded band across the whole sweep. Drift out of band indicates the loop is failing at that parameter value.

## Spectral radius

*What it is:* The largest-magnitude eigenvalue of the recurrent weight matrix  $J$ . A linear stability indicator: if  $\rho(J) > 1$ , the linearised network would explode;  $\rho(J) \approx 1$  sits at the edge of chaos.

*How it's calculated:* The recurrent matrix is reassembled post-training with E columns kept positive and I columns flipped to negative (so Dale's law appears as signs), then  $\rho(J) = \max_i |\lambda_i|$  via an eigendecomposition.

*Common values:*  $\rho < 1$  is quiescent (no sustained oscillation);  $\rho \approx 1$  is the gamma operating regime;  $\rho > 1$  is linearly unstable but the spiking nonlinearity can still stabilise it.

## Linear $f_0$ prediction

*What it is:* A continuous-rate prediction for the oscillation frequency, derived from the eigenvalue of  $J$  with the largest imaginary part.

*How it's calculated:* Pick  $\lambda^* = a + bi$  with largest  $|b|$ , then

$$f_0^{\text{lin}} = \frac{|b|}{2\pi} \cdot 1000 \text{ Hz}$$

*Common values:* Typically an upper bound on the measured  $f_0$  — the linear mode ignores the spiking nonlinearity and the refractory floor.

## Refractory-floor prediction

*What it is:* A lower bound on the oscillation frequency set by the biology: no spiking network can cycle faster than one excitatory refractory period plus one GABA-synapse decay.

*How it's calculated:*

$$f_0^{\text{cor}} = \frac{1000}{\tau_{\text{ref}}^E + \tau_{\text{GABA}}} \text{ Hz}$$

*Common values:* The measured  $f_0$  typically lies between  $f_0^{\text{cor}}$  and  $f_0^{\text{lin}}$ .

## loss

*What it is:* The training-set cross-entropy at the end of each epoch — the scalar the optimiser is minimising.

*Common values:* Starts near  $\ln(10) \approx 2.3$  for chance-level classification on MNIST and drops monotonically under healthy training. A flat or climbing loss means the gradient isn't reaching the readout.

## test\_loss

*What it is:* Cross-entropy on the held-out test set, evaluated once per epoch.

*Common values:* Tracks *loss* with a small gap. A widening gap usually means overfitting; a gap that collapses to zero often means data leakage between train and test.

## **lr**

*What it is:* The current learning rate — the multiplier on each gradient step. Varies if `adaptive_lr` is set; constant otherwise.

*Common values:*  $10^{-2}$  for `snnTorch` / CUBA;  $10^{-4}$  for COBA / PING, whose conductance-based gradients are much steeper (see Gradient Stabilisation).

## **grad\_norm**

*What it is:* The mean gradient norm across all parameters, taken after gradient clipping.

*Common values:* Stable to within an order of magnitude across a run. A sudden jump usually precedes an NaN loss.

## **grad\_ratios**

*What it is:* A dictionary mapping each weight matrix to its  $\|\nabla\| / \|W\|$  — the quantity plotted by `grad_flow`.

*Common values:* Healthy band  $10^{-3}$  to  $10^{-2}$  per layer; below is vanishing, above is exploding.

## **acc**

*What it is:* Test-set accuracy (%) at the end of each epoch — the canonical “how well is the network classifying” number.

*Common values:* 10 % is chance on MNIST; 85–90 % is typical at a small training tier; 95–98 % at medium or larger; anything above 98 % flags a leaked test split or a bug.

## **best\_acc**

*What it is:* The maximum of `acc` over all epochs seen so far. Tracked alongside `best_epoch` (the epoch that set it) and a `new_best` flag (true on the epoch that set it).

*Common values:* This is the headline number cited in notebook entries. If `best_epoch` is near the final epoch the run was still improving; if much earlier, it had plateaued.

On inference runs, `best_acc` is just the single-shot test accuracy; the raw count `n_correct` is saved alongside, and per-sample calls go to `test_predictions.json` (one row per sample, carrying `idx`, `true`, `pred`, `correct`, `logits` for confusion-matrix and calibration downstream).